

WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-06-22 to 2026-06-28

Curated articles in political communication and computational social science.

SUMMARY

COMAP Journal Bot reviewed 246 collected candidates for the 2026-06-22 to 2026-06-28 digest window. After manual relevance filtering, the curated digest keeps 30 papers: 15 new this week, 15 previous-week catch-up items, and 0 late additions.

This week is especially strong for computational media and political institutions. The new papers connect LLM-based measurement of foreign-news coverage, podcast audience epistemics, social-media negativity, workplace political segregation, and visual cues in elite selection. The policy-facing side is equally useful: police legitimacy in Mali, fair-workweek regulation, design-based causal inference, labor-managed firms, and science-agency politicization all speak to how institutions earn, organize, and lose trust.

- AI-for-CSS appears in a practical LLM news-measurement pipeline and in catch-up work on LLM agents, games, and medical AI autonomy.
- Political communication is anchored by podcasts, corrosive institutional frames, foreign-news measurement, and social-media discourse dynamics.
- Governance and inequality threads run through policing, labor scheduling, workplace segregation, polarization outside high-income democracies, and political screening in science funding.

HIGHLIGHTS

NEClass: a lightweight LLM pipeline for measuring Foreign news coverage

COMMUNICATION METHODS AND MEASURES · 2026-06-24

Tobias Schmidt

Why it matters · A direct methods contribution for computational communication: it uses an LLM-supported pipeline to classify country references in news at scale while reporting validation against stronger baselines.

[Open article →](#)

The Joe Rogan Experience: Epistemic Style Over Ideological Extremity in the Manosphere

POLITICAL COMMUNICATION · 2026-06-24

Dominik A. Stecua; Dannagal G. Young; Michael W. Wagner; Dhavan V. Shah

Why it matters · This brings podcast audiences into political communication research, linking JRE listening to epistemic orientations, media diet, conspiracism, trust, and masculinity-coded identity rather than simple party labels.

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Portraits of Power: Facial Appearance and the Tacit Domain of Political Selection in China

AMERICAN POLITICAL SCIENCE REVIEW · 2026-06-22

JUNYAN JIANG; SONGPO YANG

Why it matters · The paper uses AI-scaled facial perception measures to study promotion and purge risk among Chinese officials, connecting visual politics, authoritarian selection, and institutional power.

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Political polarization in low- and middle-income countries

NATURE HUMAN BEHAVIOUR · 2026-06-22

Horacio Larreguy; Ernesto Tiburcio

Why it matters · A useful corrective to US- and Europe-heavy polarization research, moving comparative polarization questions into lower- and middle-income contexts.

[Open article →](#)

Political segregation in the US workplace

NATURE HUMAN BEHAVIOUR · 2026-06-22

Justin Frake; Reuben Hurst; Max Kagan

Why it matters · This extends political sorting beyond neighborhoods and media feeds into workplaces, a social setting where repeated contact may shape political exposure and civic life.

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Differentiation drives the erosion of positivity on social media

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-06-22

Hongkai Mao; Yuan Chang Leong; Yutong Jiang; Alex Koch; William J. Brady; Joshua Conrad Jackson

Why it matters · The paper offers a mechanism for why online conversations become more negative even without a taste for negativity, backed by a very large Reddit corpus.

[Open article →](#)

Fair Workweek laws in the US: An appraisal of intended and unintended consequences

SCIENCE ADVANCES · 2026-06-26

Daniel Schneider; Kristen Harknett; David Arbelaez

Why it matters · A policy-facing labor study with original survey data and difference-in-differences evidence on scheduling regulation, enforcement, and job-quality tradeoffs.

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Rebuilding trust in national police: The case of the UN mission in Mali

AMERICAN JOURNAL OF POLITICAL SCIENCE · 2026-06-25

Nadine Ansorg; Zorzeta Bakaki; Jessica Di Salvatore

Why it matters · It connects peacekeeping deployment, Afrobarometer survey evidence, and public trust in national police, making it useful for governance and institutional legitimacy research.

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Previous Week Catch-Up

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COLLECTION SNAPSHOT		
TOTAL COLLECTED CANDIDATES 246	NEW THIS WEEK CANDIDATES 89	PREVIOUS WEEK CATCH-UP CANDIDATES 157
LATE ADDITIONS CANDIDATES 0	CURATED DIGEST BELOW 15 new, 15 catch-up, 0 late additions	NOTE The full collected set above is before manual relevance filtering, while the HTML email body, attached full HTML artifact, and attached PDF reflect the curated digest.

FULL CURATED DIGEST

New This Week

Nature

Inside the new political screening that's stalling NIH grants

2026-06-26 · DOI 10.1038/D41586-026-01924-8

*Max Kozlov***Abstract.** Abstract unavailable.[Read article →](#)

Science Advances

Fair Workweek laws in the US: An appraisal of intended and unintended consequences

2026-06-26 · DOI 10.1126/SCIADV.AEA8632

Daniel Schneider; Kristen Harknett; David Arbelaez

Harvard Kennedy School, Cambridge, MA, USA.; University of California, San Francisco, San Francisco, CA, USA.

Abstract. Fair Workweek laws, implemented in several large cities and one state, represent a new wave of innovation in a century of contestation and legislation around work time. Where previous legislation sought to regulate overwork, these laws aim to regulate work schedule instability and insufficiency of work hours, but evidence on their effects is incomplete. Using stacked difference-in-differences models and original survey data, we find robust evidence that Fair Workweek regulations led to increases in advance notice of schedules and time to rest between shifts and reductions in last-minute schedule changes. We find no evidence of compensatory retrenchment in other aspects of job quality. These average effects disguise substantial heterogeneity across jurisdictions, pointing to the importance of local enforcement in realizing policy goals. Fair Workweek legislation has realized some intended effects on work schedules without unintended consequences, but the potential for more widespread, effective enforcement has not been fully realized.

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American Journal of Political Science

Rebuilding trust in national police: The case of the UN mission in Mali

2026-06-25 · DOI 10.1111/AJPS.70066

Nadine Ansorg; Zorzeta Bakaki; Jessica Di Salvatore

Associate Professor, Department of Government University of Essex Colchester UK; Associate Professor, Department of Political and Social Sciences University of Florence Florence Italy; Reader, Politics at Kent University of Kent Canterbury Kent UK

Abstract. Abstract International interventions often aim to reinforce both capacity as well as perceived legitimacy of national security forces. However, how peacekeeping operations manage to improve trust in the national police has received limited attention. In this article, we evaluate whether and how UN missions can (re)build trust in the national police focusing on the case of Mali. We combine data on UN local deployment and nationally representative survey rounds from the Afrobarometer. We use the wealth of information from the survey to disentangle the mechanisms linking UN peacekeeping to positive changes in trust in national police. Employing a difference-in-difference estimation, we find that respondents in locations that

hosted UN police report higher trust toward the national police forces compared to respondents in the control group. We find that this may be due to UN police improving perceptions of safety, accessibility to and integrity of national police forces.

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Political Analysis

On the Foundations of the Design-Based Approach

2026-06-25 · DOI 10.1017/PAN.2026.10045

P. M. Aronow; Austin Jang; Molly Offer-Westort

University of Chicago Division of the Social Sciences; Yale University

Abstract. Abstract The design-based paradigm may be adopted in causal inference and survey sampling when we assume Rubin's stable unit treatment value assumption (SUTVA) or impose similar frameworks. While often taken for granted, such assumptions entail strong claims about the data-generating process. We develop an alternative design-based approach: we first invoke a generalized, non-parametric model that allows for unrestricted forms of interference, such as spillover. We define an associated set of inferential targets and discuss their interpretation under SUTVA and a weaker assumption that we call the "no unmodeled revealable variation assumption" (NURVA). We then reconstruct the standard paradigm, reconsidering SUTVA at the end rather than assuming it at the beginning. Despite its similarity to SUTVA, we demonstrate the practical limitations of NURVA alone for identifying substantively interesting quantities. In so doing, we provide clarity on the nature and importance of SUTVA for applied research.

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Communication Methods and Measures

NEClass: a lightweight LLM pipeline for measuring Foreign news coverage

2026-06-24 · DOI 10.1080/19312458.2026.2692353

Tobias Schmidt

TU Dortmund University

Abstract. Understanding cross-national news coverage is central to international communication research. However, quantifying how countries appear in the news remains methodologically challenging: simple keyword counts lack context, while manual coding does not scale. This paper introduces NEClass, a reproducible, open-source pipeline that combines standard Named Entity Recognition (NER) with a fine-tuned Large Language Model (LLM) to classify entities' country affiliations. Unlike static dictionary approaches, NEClass leverages the latent world knowledge of modern LLMs and local context to resolve ambiguities dynamically. I demonstrate the pipeline's validity regarding country attribution (weighted F1 of 0.95), show that it performs more reliably than established baseline approaches (including keyword matching and zero-shot LLM prompting), and illustrate its utility through a case study of the Eurozone crisis coverage in German and Italian newspapers. Ultimately, NEClass provides researchers with a robust and accessible toolkit to operationalize international attention patterns beyond the limitations of purely keyword-based metrics.

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Political Communication

The Joe Rogan Experience: Epistemic Style Over Ideological Extremity in the Manosphere

2026-06-24 · DOI 10.1080/10584609.2026.2690998

Dominik A. Stecua; Dannagal G. Young; Michael W. Wagner; Dhavan V. Shah

The Ohio State University; University of Delaware; University of Wisconsin-Madison

Abstract. Despite increasing prominence in the contemporary media ecosystem, podcasts have attracted limited scholarly attention. This study examines audience characteristics of the most popular podcast in the United States, The Joe Rogan Experience (JRE), situating it within the broader cultural context of the "manosphere." Leveraging two national probability samples collected in October 2022 (N = 2,907) and November 2023 (N = 1,000), this research explores the socio-demographic profile, information environment, epistemic orientations, identity, ideology, and trust of Rogan's listeners. Across both studies, among the strongest correlates of JRE listening is conservative media consumption, with additional associations with skeptical epistemic orientations, conspiracism, political and institutional distrust, and being young and male. In contrast, partisanship is not significantly associated with JRE listening once these other factors are included in the models. Contrary to common portrayals, hostile sexism, authoritarianism, and racial attitudes do not reliably predict JRE listening once media diet is accounted for. This research underscores the importance of examining audiences' deeper cultural values and epistemic orientations alongside traditional political variables, thereby clarifying our understanding of how influential media figures such as Rogan shape and reflect broader currents in an increasingly fragmented information environment.

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American Political Science Review

Efficient Production without Domination: The Case of Labor-Managed Firms

2026-06-23 · DOI 10.1017/S0003055426101750

INIGO GONZALEZ-RICOY

University of Barcelona

Abstract. Labor-managed firms, whether in the form of worker cooperatives, codetermination, or broad-based employee ownership, have recently prompted substantial public debate, to which political theorists have contributed with normative analyses that are seriously deficient insofar as they have neglected considerations of productive efficiency. I offer a normative view of labor-managed firms that leans squarely on efficiency considerations. I first argue that the value of productive efficiency is grounded not only on the interests of capital suppliers and society at large, but also on workers' interest in avoiding domination by their bosses, and examine how efficiency may serve and set back this interest. I next argue that some variants of labor-managed firms can secure efficient production while protecting workers' interest in nondomination more robustly than capital-controlled firms. Finally, I compare labor-managed firms to alternatives to achieve these aims, including self-employment, workplace regulation, unionization, and labor market policies to enhance workers' exit options.

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Portraits of Power: Facial Appearance and the Tacit Domain of Political Selection in China

2026-06-22 · DOI 10.1017/S0003055426101749

JUNYAN JIANG; SONGPO YANG

Columbia University; Peking University

Abstract. As the epitome of modern, rational organizations, bureaucracies are often believed to select candidates based on rules and reason. We argue that intuitive-and even instinctive-assessments of candidates' external appearances sometimes underpin seemingly rational and calculated decisions. Using a novel, AI-based algorithm that learns and reproduces human assessments of facial appearances at scale, we examine how perceived facial traits influence the careers of over four thousand mid- and senior-level Chinese officials. We find that officials who look more competent, trustworthy, and less aggressive enjoy significantly better promotion prospects and lower purge risk than their peers. Warmth-related traits (e.g., trustworthiness and non-aggressiveness) are especially valued at higher-level promotions and for male candidates. Additional analyses, including conjoint experiments with real officials, demonstrate that appearances' influence over selection

preferences is comparable to performance or political connections. These findings challenge the prevailing meritocratic and relation-based theories of bureaucratic selection and highlight the role of impressions in the workings of government institutions.

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Communication Research

Fifty Years of Diversity in Communication and Media Studies (1974-2023): The Intertwined Trends in Geographic, Gender, and Methodological Representation

2026-06-22 · DOI 10.1177/00936502261454813

Mark Boukes; Irina Lock; Shujun Liu

Cardiff University, UK; Friedrich Schiller University Jena, Germany; University of Amsterdam, The Netherlands

Abstract. This study explored trends of diversity in Communication and Media Studies (CMS) between 1974 and 2023 through an automated content analysis of journal publications. An increasing proportion of first-authors from non-WEIRD countries and non-U.S. WEIRD countries was found alongside a declining proportion of U.S.-based scholars (WEIRD refers to "Western, Educated, Industrialized, Rich, and Democratic"). The visibility of women first-authors grew markedly to 54% in the 2020s, although still a bit below the benchmark. A growing prevalence of qualitative methods was found relative to quantitative methods and conceptual work. Interestingly, these trends were intertwined. Qualitative studies were more likely to be authored by women, by scholars from non-U.S. WEIRD countries, and based on non-WEIRD populations. Women authorship was less likely among the increasingly visible non-WEIRD and non-U.S. WEIRD first-authors. Furthermore, less diversity-in particular, geographic and methodological diversity (but not gender diversity)-was found in journals with higher impact-factor.

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Nature Human Behaviour

Political polarization in low- and middle-income countries

2026-06-22 · DOI 10.1038/S41562-026-02507-3

Horacio Larreguy; Ernesto Tiburcio

Abstract. Abstract unavailable.

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Political segregation in the US workplace

2026-06-22 · DOI 10.1038/S41562-026-02501-9

Justin Frake; Reuben Hurst; Max Kagan

Abstract. Abstract unavailable.

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Proceedings of the National Academy of Sciences

Differentiation drives the erosion of positivity on social media

2026-06-22 · DOI 10.1073/PNAS.2527316123

Hongkai Mao; Yuan Chang Leong; Yutong Jiang; Alex Koch; William J. Brady; Joshua Conrad Jackson

Behavioral Science Department, Booth School of Business, University of Chicago; Data Science Institute, University of Chicago; Department of Management and Organizations, Kellogg School of Management, Northwestern University; Department of Psychology, Harvard University; Department of Psychology, University of Chicago; Organizational Behavior Department, Graduate School of Business, Stanford University

Abstract. Most people believe that social media discourse is negative and divisive. Here we show how this negativity can evolve even when users are not motivated to be negative. We propose that social media users seek to differentiate themselves from other users, and it is easier to differentiate oneself through negativity than positivity because negative information is more heterogeneous and counternormative than positive information. This makes users increasingly likely to post negative comments as a conversation unfolds and it becomes more challenging to make unique contributions. Analyzing 2.05 billion comments from 2,150 Reddit communities shows that comments become more negative over time, both within threads and community histories. This trend toward negativity is mediated by the semantic uniqueness of comments, suggesting that it arises from users differentiating themselves. This trend is strongest when initial dialogue is positive, making negative comments highly counternormative. We replicate these patterns in an experiment simulating social media dialogue ($n = 3,685$). Participants become more negative over time, but only when incentivized to be unique, and especially when dialogue begins positively. These findings suggest that the structure of social media platforms interacts with human motivation to foster a drift toward negativity over time in online discourse.

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People prefer to negotiate with women, even when outcomes are identical and gender is unknown

2026-06-22 · DOI 10.1073/PNAS.2523202123

Charlotte H. Townsend; Laura J. Kray; Solene Delecourt

Haas School of Business, University of California Berkeley; Industrial & Labor Relations School, Cornell University

Abstract. While gender and economic outcomes in negotiation have been studied for decades, much less is known about gender differences in subjective outcomes, such as trust, rapport, and willingness to negotiate again, despite their importance for long-term success. In a series of studies with over 2,000 participants, we find that people consistently report better subjective outcomes when negotiating with women. This preference persists even in anonymous negotiations where gender is unknown and cannot be inferred from behavior, as well as in conditions where negotiation partner gender is randomly assigned. Importantly, we find no gender difference in economic outcomes. To understand why, we analyzed transcripts of real negotiations for any behavioral differences. These findings reveal an overlooked gender dynamic: While men and women achieve similar economic results, women foster stronger interpersonal relationships, which in turn lead to greater satisfaction and greater desire for future negotiations with women.

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Three centuries of technological innovation: Opportunity is the mother of invention

2026-06-23 · DOI 10.1073/PNAS.2525310123

Jordan G. Okie; James H. Brown; Astrid Kodric-Brown; Joseph R. Burger; Tatiana P. Flanagan; Trevor S. Fristoe; Sean T. Hammond; Norman Mercado-Silva; Jeffrey C. Nekola; Jennifer Richter

Centro de Investigacion en Biodiversidad y Conservacion, Universidad Autonoma del Estado de Morelos, Chamilpa; Department of Biology, University of Kentucky; Department of Biology, University of New Mexico; Department of Biology, University of Puerto Rico-Rio Piedras; Department of Botany and Zoology, Faculty of Science, Masaryk University; Department of Earth System Science and Policy, University of North Dakota; School for the Future of Innovation in Society, Arizona State University; School of Earth and Space Exploration, Arizona State University; School of Math, Science, and Engineering, Central New Mexico Community College; School of Social Transformation, Arizona State University

Abstract. What socioecological conditions nurture the ingenuity and collaborative interactions underlying transformative technological innovations? We compiled a dataset on more than 400 major technological inventions from 1690 to 1990

spanning seven categories (agriculture, armaments, information and communication, household, industry, medical, and transportation) and performed inductive macroecological analyses to address how attributes of inventors, teams, and their social and geographic environments contributed to the innovation of new technologies that have changed the way people live. The vast majority of inventions were attributed to single inventors of various ages. The frequency, size, and cultural diversity of teams increased in the 20th Century, as did the proportion of inventions attributed to women and immigrants. A wide variety of environments acted as innovation hubs, including rural areas as well as specialized institutions and large cities. Invention rate (number of inventions per time period) peaked during the 1800s in cities, rural areas, and most categories but continued increasing over time in institutional environments and medical and communication technologies. The overall pattern across ages, genders, team attributes, environments, technological categories, and time periods supports the conclusion that opportunity, ingenuity, and environment all play key roles in the inventiveness phase of the innovation process: creative individuals in particular ecological environments and social settings come up with novel solutions to specific practical problems. Similar innovation patterns have been observed for tool use among nonhuman primates, highlighting unifying processes of innovation. Our macroecological approach offers valuable insights into the emergence and drivers of innovation and material culture.

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Exploring memory effects: Sparse identification in vector-borne diseases

2026-06-25 · DOI 10.1073/PNAS.2601074123

Dimitri Breda; Muhammad Tanveer; Jianhong Wu; Xue Zhang

Department of Mathematics and Statistics, Centre of Excellence in AI for Public Health Advancement, York University; Department of Mathematics, Computer Science and Physics, Computational Dynamics Laboratory, University of Udine; Department of Mathematics, Northeastern University

Abstract. Predicting the human burden of vector-borne diseases from limited surveillance data remains a major challenge, particularly in the presence of nonlinear transmission dynamics and delayed effects arising from vector ecology and human behavior. We develop a data-driven framework based on an extension of Sparse Identification of Nonlinear Dynamics to systems with distributed memory, enabling discovery of transmission mechanisms directly from time series data. Using severe fever with thrombocytopenia syndrome as a case study, we show that this approach can uncover key features of tick-borne disease dynamics using only human incidence and local temperature data, without imposing predefined assumptions on human case reporting. We further ascertain the robustness of the recovered incidence-temperature model by integrating it with mechanistically derived tick-host covariates, showing that the forecasting ability does not improve. This suggests that the proposed core data-driven model already delivers strong predictions. The framework also allows for systematic sensitivity analysis of memory kernels and behavioral parameters. Although the approach prioritizes predictive accuracy over mechanistic transparency, it yields sparse, interpretable integral representations suitable for epidemiological forecasting. This methodology provides a scalable strategy for forecasting vector-borne disease risk and informing public health decision-making under data limitations.

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Previous Week Catch-Up

Science Advances

Heatwaves enable wildfire activity in the western United States

2026-06-19 · DOI 10.1126/SCIADV.AEA1277

Dmitri A. Kalashnikov; John T. Abatzoglou; Emily L. Williams; Cong Yin; Madhulika Gurazada; Mukesh Kumar; Ashwin P. Thomas; Precious E. Ebiende

School of Engineering, University of California, Merced, Merced, CA, USA.; Sierra Nevada Research Institute, University of California, Merced, Merced, CA, USA.

Abstract. While overall impacts of heatwaves have been extensively studied, the connection between heatwaves and wildfire activity remains relatively underexplored. We analyze links between heatwaves and both wildfire occurrence and growth across the western United States (WUS) and find that 42% of burned area during 2001-2024 occurred during and immediately following heatwaves. Heatwaves facilitate significant increases in daily burned area through meteorological and fuel flammability conditions that promote new ignitions and exacerbate ongoing fire activity, with effects persisting post-heatwave in most regions. In addition, heatwaves co-occur with increased cloud-to-ground lightning that can potentially increase ignitions. Last, we observe a 2.5-fold increase in burned area in WUS forests since 2001, with ~64% of this increase coinciding with heatwaves, but without corresponding increases in nonforests. The growing influence of heatwaves in shaping burned area in WUS forests has important implications for fire management and public health and can improve predictions of wildfire risk.

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Nature Communications

Multi-hazard exposure and vulnerability drive rising agricultural risk in global breadbasket regions

2026-06-18 · DOI 10.1038/S41467-026-74617-5

Hossein Tabari; Parisa Hosseinzadehtalaei

Abstract. Abstract Climate extremes increasingly threaten global food production, yet global multi-hazard agricultural risk assessments that jointly consider multiple hazards, crop exposure and socioeconomic vulnerability are still missing. This gap limits our ability to identify where climate shocks to croplands are most likely to translate into high agricultural risk. Here we present a global agricultural risk assessment that quantifies the exposure of croplands and ten major crop groups to heatwaves, droughts, floods and compound hot-dry events, and combines this with a vulnerability index constructed from socio-economic and institutional indicators using latent factor modelling. Using ensembles of climate and hydrological models under multiple future scenarios, we show widespread increases in multi-hazard exposure, with the strongest intensification across South Asia, Sub-Saharan Africa and major U.S. and European breadbaskets. A significant relationship between vulnerability and cropland exposure emerges in future scenarios but is absent historically, indicating a growing role for vulnerability in shaping future agricultural risk. These results demonstrate that reliably identifying future agricultural risk hotspots requires jointly accounting for multi-hazard exposure and socioeconomic vulnerability, so that adaptation efforts can be targeted to regions where intensifying extremes coincide with high vulnerability.

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Stewardship without walls: taming the unregulated antibiotic economy to contain antimicrobial resistance in Ethiopia

2026-06-18 · DOI 10.1038/S41467-026-74671-Z

Balew Arega; Asnake Agunie

Abstract. Ethiopia's unregulated private antibiotic economy spans private hospitals, clinics, pharmacies, drug shops and informal vendors. It drives resistance through three forces: intergenerational self-medication (reservoir habit), a satisfaction imperative linking sales to customer retention, and a corner drugstore calculus prioritising immediate access over cheaper but slow public care. Non-prescribed dispensing becomes the norm and selects resistant bacteria. We propose a six-pillar framework: tiered accreditation, point-of-service education, supply chain engagement, mystery client support, technology enabled consultation and operational research. This stewardship without walls is an urgent priority for Ethiopia and similar settings. Arega and Agunie analyze Ethiopia's unregulated private antibiotic economy and propose a six-pillar framework of stewardship without walls to contain antimicrobial resistance in low-income settings.

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Inequality in human development amplifies climate-related disaster risk

2026-06-17 · DOI 10.1038/S41467-026-73873-9

Khalil Teber; Sebastian Sippel; Melanie Krause; Jakob Zscheischler; Miguel D. Mahecha

Abstract. Abstract The impacts of climate-related disasters are shaped by the interaction between hazard intensity, exposure, and vulnerability. However, the influence of hazard intensity and within-country inequality on impact magnitudes remains poorly quantified. Here, we present a global multi-hazard study of over 7000 climate-related disasters reported by the Emergency Events Database from 1990 to 2020. Using subnational indicators, we show that human development drives major shifts in global exposure and impact patterns, with societal vulnerability outweighing hazard intensity in shaping impacts. Despite a declining share of global exposure over the past three decades, regions with low subnational Human Development Index scores experience disproportionately higher human losses across most disaster types. For instance, individuals in these regions face an 8.2-fold higher risk of fatality associated with storms (95% confidence interval: 2.16-23.06) compared to those in very high human development regions. Our findings also indicate that within-country inequality in human development exacerbates disaster risk in regions with low and medium levels of human development. These results underscore the critical role of human development in managing disaster risks and highlight the link between socioeconomic conditions and vulnerability to climate-related hazards.

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Is the elimination of violence against women a realistic goal for the near future?

2026-06-16 · DOI 10.1038/S41467-026-74656-Y

Enrique Gracia

Abstract. As 2030 approaches, achieving Sustainable Development Goal Target 5.2, which aims to eliminate all forms of violence against women, appears increasingly unrealistic. Using the European Union as a case study, this Perspective argues that prevalence remains high, progress has stagnated, levels can be disproportionately high in some of the most socioeconomically advanced and gender-equal countries, and prevention efforts seem unable to reach younger generations. Together, these patterns suggest that prevention and response efforts, as implemented to date, have not produced sustained population-level reductions. Achieving measurable declines will require scaling evidence-based prevention with sustained political will and stronger accountability. This Perspective argues that violence against women remains persistently high, in some of the most socioeconomically advanced and gender-equal countries as well. Progress has stagnated and prevention efforts seem unable to reach younger generations.

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Proceedings of the National Academy of Sciences

Video games help push the boundaries of AI

2026-06-18 · DOI 10.1073/PNAS.2618819123

Stephen Ornes

Abstract. In one recent project that used video games to explore artificial intelligence, computer scientists introduced 1,000 AI-driven agents into the game Minecraft and watched as an AI civilization emerged. It included artists, chefs, and explorers. Image credit: Shutterstock/ mkfilm. A few years ago, a group of computer scientists set 25 players loose in Smallville, a custom-made virtual "sandbox world"—an open-ended video game like Minecraft that allows players to freely explore and interact with each other. The catch: The players in Smallville were all virtual, each driven by its own large language model, or LLM. The goal, says Joon Sung Park, who led the project while finishing his PhD at Stanford University in California, was to see what kinds of human-like behaviors evolved. To get the players started, Park and his colleagues wrote a

paragraph describing the identity of each. For example, for one they wrote, "John Lin is a pharmacy shopkeeper at the Willow Market and Pharmacy who loves to help people." What happened next was up to the LLMs. In the spring of 2023, the researchers reported on what unfolded over two game days in Smallville (1). The game's AI populace had been surprisingly busy with familiar-seeming tasks. "They would wake up in the mornings, go to work, and do all kinds of things," Park says. When one agent was instructed (by a researcher on Park's team) to throw a Valentine's Day party with no further instructions, it issued invitations and began party preparations. Some Smallvillians began asking dates to the party and making plans to get there on time. When one agent told another that it planned to run for mayor, the campaign quickly became a hot topic in Smallville as agents discussed how they planned to vote. The sandbox environment gave Park a way to explore human behavior using LLMs; his study is one of the latest in a long history of using games to push computer technology forward. Some scientists develop games to explain difficult concepts, or even to crowdsource new ideas in a field (2). "When it comes to developing artificial intelligence, games were there from the beginning," says Julian Togelius, a computer scientist and video game researcher at New York University who explores the intersection of AI and video games. Games like chess and Go, he says, gave early researchers a way to test the capabilities of new AI tools-and remain valuable touchstones for new insights (3).

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AI agents are sensitive to nudges

2026-06-15 · DOI 10.1073/PNAS.2537030123

Manuel Cherep; Pattie Maes; Nikhil Singh

Department of Computer Science, Dartmouth College; Media Lab, Massachusetts Institute of Technology

Abstract. Large language models (LLMs) are increasingly deployed as autonomous agents that make choices and use tools on behalf of users. Yet, we have limited evidence about how their decisions are shaped by their environment. We adapt a human decision-making task to test leading LLMs under four forms of choice architecture: defaults, suggestions, information highlighting, and "optimal" nudges derived from a resource-rational model of human choice. We treat human behavior as a baseline for predictable sensitivity to such interventions. Across models and prompting strategies, LLMs often depart substantially from this baseline. They sometimes pay excessive costs to acquire information, sometimes ignore available information, and, most crucially, are far more responsive to nudges than humans, such that weak cues that slightly shift human behavior have larger effects on model choices, toward both better and worse payoff outcomes. Chain-of-thought prompting and in-context human data do not reliably stabilize behavior. Recent reasoning-optimized LLMs can, in some configurations, restore more human-level sensitivity to nudges, but do so inconsistently and at substantial computational cost. These results point to an important and largely neglected safety concern: LLM agents can be behaviorally brittle under subtle changes in choice architecture, even in the absence of adversarial settings.

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Accounting for uncertainty and bias in archaeological and historical evidence on wealth inequality

2026-06-15 · DOI 10.1073/PNAS.2514309123

Mattia Fochesato; Samuel Bowles

CORE Economics, University College London; Centre for Economic Policy Research; Department of Social and Political Sciences, Dondena, Centre for Research on Geography, Resources, Environment, Energy & Networks, and Bocconi Institute for Data Science and Analytics-Bocconi University; Santa Fe Institute

Abstract. The archaeological and historical record allows us to study the patterns of economic inequalities associated with sharply contrasting institutions such as communal property in societies without states, private property among small holders, and slavery, in economies relying on radically different technologies, based for example on human energy alone, other animal power, and carbon-based power. But the price of using prehistoric and historical data to expand the range of institutions and technologies under study is a substantial and typically unknown level of uncertainty and unrepresentativeness in the resulting estimates. We provide methods and code that quantify the degree of uncertainty, an approach that we term BRIDGE

(Bayesian-Resampling and Informed Priors with Data-driven Gini Estimation). We transform raw wealth data from 431 sites and dates into probability distributions representing likely levels of inequality that (insofar as possible) are representative of the underlying population (e.g., rectifying biases arising from small or nonrandom samples and the frequent absence of data on those entirely without wealth) and are comparable across differing asset types (e.g., burial goods, dwelling area, storage area, land cultivated), ownership units (individual or household), and scale (from villages to nations). These distributions are based on the raw data along with independent information that allows us to infer biases and levels of uncertainty. We also account for the propagation of uncertainty through every stage of the data generation and estimation process. We find that widely used methods (e.g., relying solely on conventional bootstrapping) provide misleading and for the most part substantially underestimated measures of uncertainty.

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Oxytocin modulates the neurocomputational mechanisms engaged in learning rank relationships in social networks

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Jiawei Liu; Chen Qu; Remi Philippe; Siying Li; Edmund Derrington; Jean-Claude Dreher

Faculty of Education, Northeast Normal University; Key Laboratory of Brain, Cognition and Education Sciences (South China Normal University), Ministry of Education; Laboratory of Neuroeconomics, Institut des Sciences Cognitives Marc Jeannerod; School of Psychology, Center for Studies of Psychological Application, and Guangdong Key Laboratory of Mental Health and Cognitive Science, South China Normal University; Universite Claude Bernard Lyon 1

Abstract. Indirect evidence from preclinical and neuropharmacological studies suggests that oxytocin may attenuate social dominance relationships and modulate social memory. In humans, oxytocin may also modulate self- versus other-oriented reward learning and in-group/out-group decisions differently. Although the neural bases of learning social hierarchy by observation have been identified for linear hierarchies (a 1 2 n), the brain mechanisms engaged in learning more complex rank relationships (i.e.) in social networks remain unknown. Here, we investigated the modulatory role of oxytocin on the neurocomputational mechanisms engaged when learning ranks in social networks and when making transitive inferences based on social memory retrieval. Participants learned rank relationships between members of two social networks, one in which they were embedded and the other not. During a subsequent test phase, they inferred rank relationships between pairs of members not encountered before. During training, oxytocin (vs. placebo) improved choice accuracy regarding rank relationship between pairs of members for other-oriented network and increased activity of the ventromedial prefrontal cortex. During the test phase, oxytocin modulated the balance between self/other social memory, boosting performance and amygdala responses for memory retrieval concerning social networks in which participants were not embedded. These findings demonstrate that oxytocin modulates the brain computations needed to learn and retrieve rank relationships in social networks in a manner that depends upon reference-frame.

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Science

Changes in wildlife activity patterns in response to war in Ukraine

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Svitlana Kudrenko; Richard Bischof; Andreas Zedrosser; Nuria Selva; Kateryna Korepanova; Denys Vyshnevskiy; Martin Gahbauer; Serhii Obrizan; Serhii Domashevskiy; Oleksandr Borsuk; Alona Varukha; Marco Heurich

Bavarian Forest National Park, Grafenau, Germany.; Chornobyl Radiation and Ecological Biosphere Reserve, Ivankiv, Kyiv region, Ukraine.; Departamento de Ciencias Integradas, Facultad de Ciencias Experimentales, Centro de Estudios Avanzados en Fisica, Matematicas y Computacion, Universidad de Huelva, Huelva, Spain.; Department for Integrated Biology, Institute of Wildlife Biology and Game Management, University for Natural Resources and Life Sciences, Vienna, Austria.; Estacion Biologica de Donana Consejo Superior de Investigaciones Cientificas, Sevilla, Spain.; Faculty of Environment and Natural Resources, University of Freiburg, Freiburg im Breisgau, Germany.; Faculty of Environmental Sciences and Natural Resource Management, Norwegian University of Life Sciences, As, Norway.;

Faculty of Technology, Natural Sciences and Maritime Sciences, Department of Natural Sciences and Environmental Health, University of South-Eastern Norway, B, Norway.; Frankfurt Zoological Society, Frankfurt, Germany.; Graduate School of Symbiotic Systems Science and Technology, Fukushima University, Fukushima-shi, Japan.; Institute for Forest and Wildlife Management, University of Inland Norway, Koppang, Norway.; Institute of Nature Conservation, Polish Academy of Sciences, Krakow, Poland.; Sector of Sustainable Development and Environmental Assessment, Institute of Geography, National Academy of Sciences of Ukraine, Kyiv, Ukraine.

Abstract. Conflict zones are inherently hazardous and inaccessible for researchers, which results in a knowledge gap about the immediate effects of armed conflicts on the environment, particularly wildlife. We used camera-trap detections to investigate the impact of armed conflict on wildlife activity patterns before, during, and after the Russian occupation of the Chernobyl Exclusion Zone (Ukraine) in 2022 and compared it to the same period in 2021. Mammal species responded to armed conflict through immediate behavioral adjustments, including reduced activity during night and on dates when armed-conflict activities intensified. Our results provide insight into wildlife's behavioral responses to armed conflict in real time and underscore the potential of camera trapping to quantify the ecological effects of war.

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New list of political jobs at science agencies called 'weird and capricious'

2026-06-18 · DOI 10.1126/SCIENCE.AEJ7945

Jeffrey Mervis

Abstract. NIH programs and leadership are hit hard; other science agencies are barely mentioned

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American Political Science Review

Beyond Belief Change: The Persuasive Returns of Targeting Attitude-Relevant Beliefs

2026-06-17 · DOI 10.1017/S0003055426101622

YAMIL RICARDO VELEZ; PATRICK LIU; SCOTT CLIFFORD

Columbia University; Texas A&M University

Abstract. A persistent puzzle in the study of public opinion is why political information often produces minimal attitude change despite reliably influencing beliefs. We argue that this duality reflects belief relevance—the extent to which specific beliefs bear on attitudes. Using conversations with large language models (LLMs), we elicit deeply held issue attitudes and the “focal beliefs” people use to justify those attitudes. We then randomly assign participants to receive an LLM-generated counterargument targeting either their focal belief, an issue-relevant but unmentioned belief (“distal belief”), or a placebo. In experiments with two large online convenience samples, counterarguments targeting the aforementioned beliefs successfully decrease belief strength, with effects persisting after 10 days. More importantly, focal belief counterarguments produce larger and more durable attitude change than distal counterarguments. These findings suggest that political information can successfully shift political attitudes and provide new evidence for the role of belief relevance in persuasion.

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Vernacular Architecture and Grassroots Urban Politics: How Politics Is Embedded in Residential Design

2026-06-17 · DOI 10.1017/S0003055426101592

PAIGE BOLLEN; NOAH L. NATHAN

Massachusetts Institute of Technology; The Ohio State University

Abstract. The physical structures in which urban life occurs are an underappreciated determinant of how grassroots urban politics unfolds. In many rapidly growing cities, housing scarcity forces residents into multifamily buildings that create daily

exposure to neighbors. We argue that these exposures affect political behavior by shaping residents' access to political information and capacity for collective action. We focus on the informal, vernacular architecture of West Africa's dominant urban housing form—the compound house. Compound house residents in urban Ghana participate more in politics than similar residents of other housing types. Leveraging an original survey, including novel measures of tenants' spatial network centrality within their residential buildings, we suggest that key mechanisms for this relationship emerge from the effects of architectural design on visibility and social ties among co-tenants. Ultimately, built environments must be studied alongside demographic environments to best understand contextual effects on political behavior.

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Political Communication

Democratic Institutions Under Threat: Corrosive Frames, Internet News, and the U.S. Supreme Court

2026-06-17 · DOI 10.1080/10584609.2026.2682952

Justin Wedeking; Michael A. Zilis

University of Kentucky

Abstract. Are democratic institutions under threat in the United States? We examine an understudied democratic bedrock: the judiciary and, in particular, the U.S. Supreme Court. We draw on data from "SCOTUSapp," a tool that provides a database of 100,000+ news articles across thousands of unexamined sources. We find a new media environment where the Court receives attention from thousands of ideologically fragmented outlets. Additionally, some framings of the Court are potentially corrosive. We then use two experiments to examine the effects of these corrosive news frames. The first experiment (October 2021) provides evidence that frames portraying the Court as embroiled in secrecy, hypocrisy, illegitimacy, and politics cause a decrease in support for judicial independence. These frames are influential among Republicans and Democrats. The second experiment (May 2023) follows *Dobbs*, which severely damaged judicial faith among Democrats, and shows that corrosive frames lead to a corresponding decline only among strong Republicans.

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Nature Human Behaviour

The interaction of meaning similarity and confusability explains regularity in form-meaning mappings at and below the word level

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Thomas Brochhagen; Xixian Liao; Jamie D. Wright; Carmen Saldana

Abstract. Languages exhibit striking regularities in how meanings are mapped to word forms, yet analogous patterns at the subword level remain under-explored. This study presents a large-scale cross-linguistic analysis of regularity at and below the word level, drawing on data from over 1,900 languages. Here we find that while the co-expression of meanings at both levels is highly systematic, the meanings recurrently involved in the two levels differ. Nevertheless, regularity at and below the word level is explained by the same underlying principle: a tension between pressure for lexical compression and lexical differentiation. The former favours reusing forms for similar meanings to ease learning and processing. The latter favours the use of distinct forms to prevent ambiguity. These findings offer a unified account of lexical organization across the world's languages, with subword-level form reuse emerging as a principled compromise when full word-level reuse risks miscommunication. Languages exhibit striking regularities in how meanings are mapped to word forms, yet analogous patterns at the subword level remain under-explored. This study fills the gap with a large-scale cross-linguistic analysis of regularity at and below the word level, drawing on data from over 1,900 languages.

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